RF High Q Head RF Output Gating

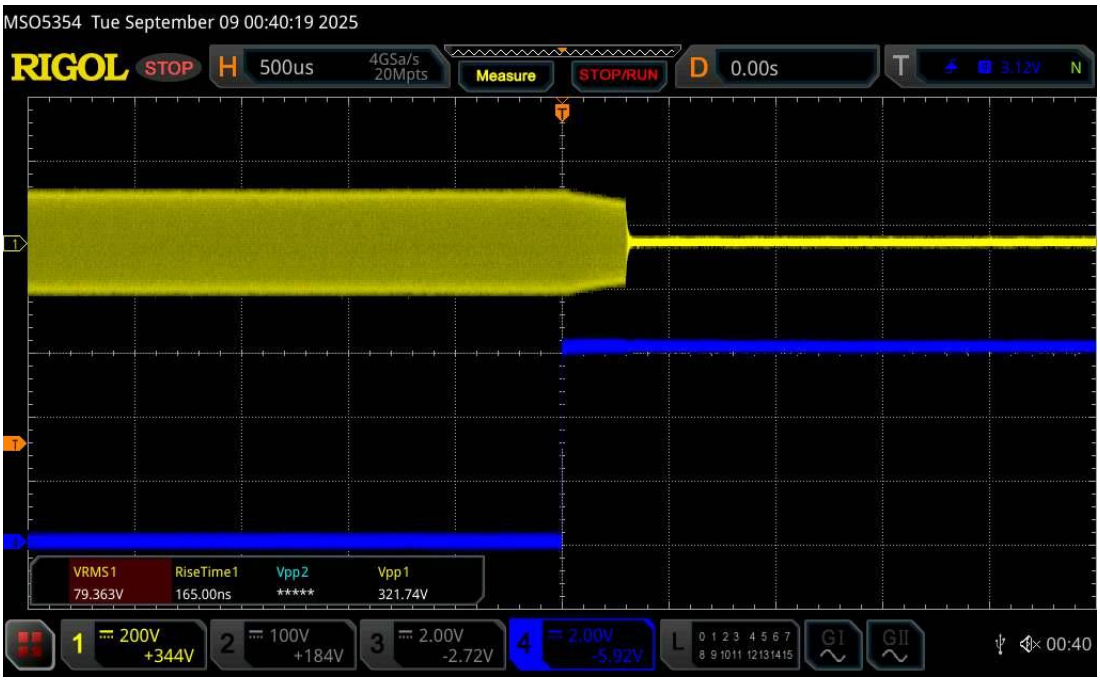
September 28, 2025

The MIPS system supports the generation of RF voltage to control devices such as RF ion guides and ion funnels. The MIPS control chassis houses RF driver modules that connect to external RF high-Q heads. These RF heads are connected using 15-pin D cables provided with the MIPS system. Each RF driver module can control two separate RF high-Q heads, a total of two RF drivers to be installed in a single MIPS system, enabling control of four RF high-Q heads.

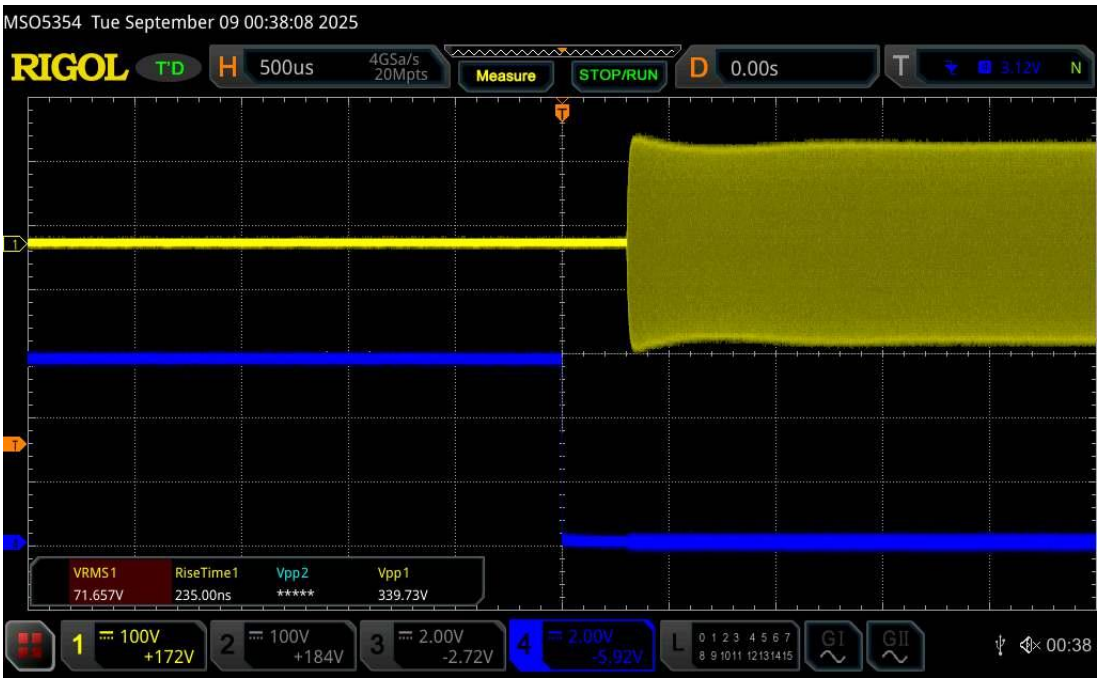
The MIPS system empowers users to manage the RF high-Q head, including monitoring and controlling the RF level. This document delves into the process of gating the RF on and off using an external TTL-level logic control signal. Gating can be accomplished by transmitting a digital logic level signal to the MIPS system, which subsequently activates and deactivates the RF driver modules' generation of the control frequency and drive control. Alternatively, the RF high-Q head can be equipped to accept a control signal. Both methods are detailed in this document.

MIPS control gating option:

The MIPS control chassis's user interface offers menu options that enable users to select an input digital signal to control the gating and define the active level of this control signal. The scope images below illustrate the typical turn-on and turn-off performance achieved using this interface.



This image depicts the gate logic level signal (blue) employed to control the RF output of a high-Q RF head. The RF output is reduced to zero between 250 nanoseconds and 1.5 microseconds after the falling edge of the gate signal. The RF signal is represented in yellow.



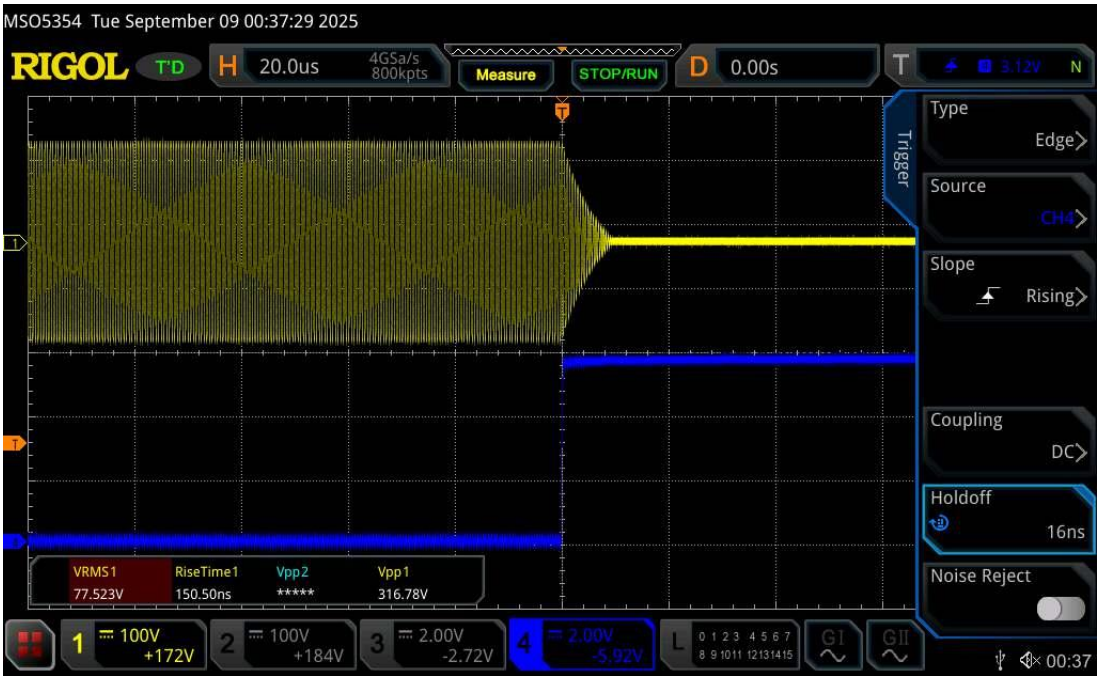
This image illustrates the RF level being gated back on. When the blue logic control signal goes low, the RF output (yellow) is enabled. The RF output level is generated between 250 nanoseconds and 1.5 microseconds after the falling edge of the gate signal.

Two models of the RF driver, internal to the MIPS controller, can be utilized in your system. MIPS systems equipped with ARB modules and more than two RF high-Q head connections (more than one RF driver module) employ a version of the RF driver that disables the MIPS-based gating options. Supporting the MIPS-based gating requires a firmware update that is currently in development and will be released in early 2026.

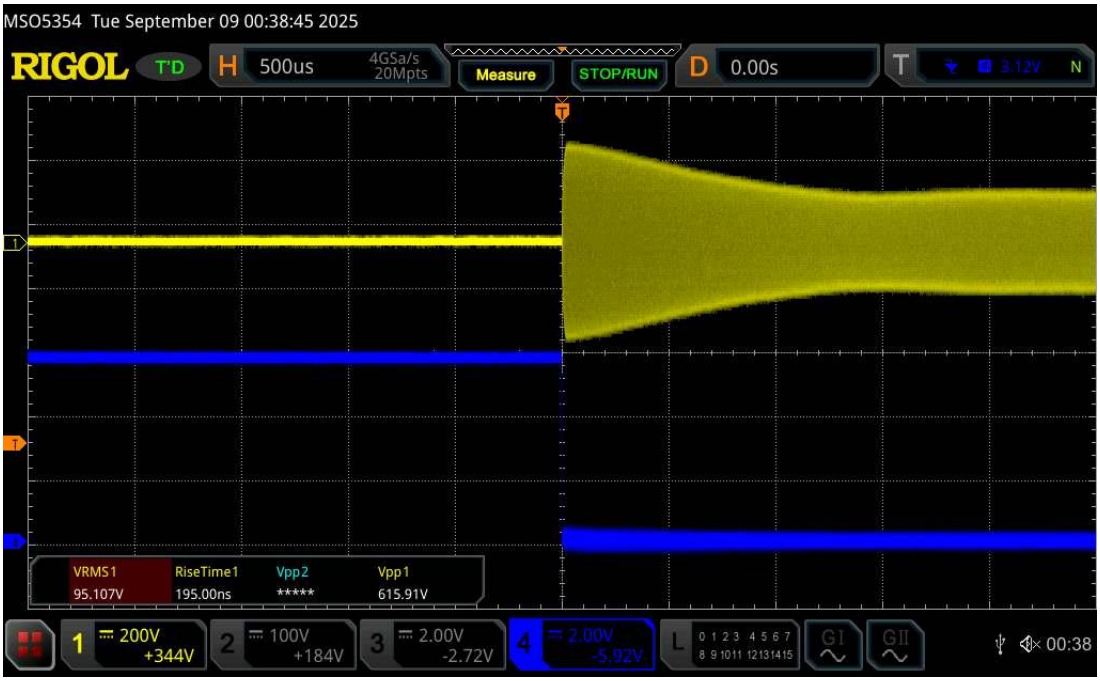
RF high-Q head gating:

RF high-Q heads can be fabricated with a high-speed gating option. In this case, the control logic level signal to enable and disable RF generation is applied to the RF head using its Gate input BNC connector. By using this gate input BNC, the RF generation is enabled and disabled in hardware, resulting in a very fast response to the control signal. The scope images below demonstrate the response time to both turning off and turning on the RF output. When the output is disabled, a ring-down occurs as the energy in the resonant circuit dissipates. In the example shown, this time is 10 microseconds, but it will vary depending on the Q of your system.

When the RF is reenabled, the RF level may exceed your desired level as the resonant circuit rings up. This depends on the Q of your system and can be adjusted by making small changes to the operating frequency of the RF head.



This image illustrates the logic level, depicted in blue, which controls the RF output of an RF high Q head. It also shows the RF output ring-down, which occurs as the energy stored in the resonant circuit is dissipated.



This image illustrates the logic level signal (in blue) that enables or gates the RF output from a high-Q RF head. As the resonant circuit rings up, you can observe the RF level overshoot. The response can be tuned by making minor adjustments to the operating frequency.

The active level of the logic level control signal can be adjusted using jumpers on the RF high-Q head. However, this requires removing the RF head from the chassis and moving the jumpers on the internal PCB. The active level is set in the factor, so that a logic 1 will disable the RF level. For detailed instructions on changing this active level, please contact the factory.